



L4 Enterprise Data Management

Business Analytics, 2e

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Learning Objectives (LOs)

Enterprise data management:

- File
- Relational Database
- Datawarehouse and Data lake
- Big data
- Embedded vectors for LLM

1.2: Data Management

- An important first step for making decisions is to find the right **data** and prepare it.
 - Compilation of facts, figures, or other content
 - Numerical and non-numerical
 - All types and formats are generated from multiple sources
- Data that have been organized, analyzed, and processed in a meaningful and purposeful way become **information**.
 - Often we have a large amount of data
 - Even small data can give insights

Enterprise Data Management

- What is Enterprise Data Management?
 - Definition: Managing, integrating, and governing data across an organization.
 - Importance: Drives decision-making, compliance, and innovation.
- Why Does It Matter?
 - Data is a critical asset in the digital age.
 - Poor management leads to inefficiencies, risks, and missed opportunities.

THE DATA HIERARCHY

A computer system organizes data in a hierarchy that starts with the **bit**, which represents either a 0 or a 1.

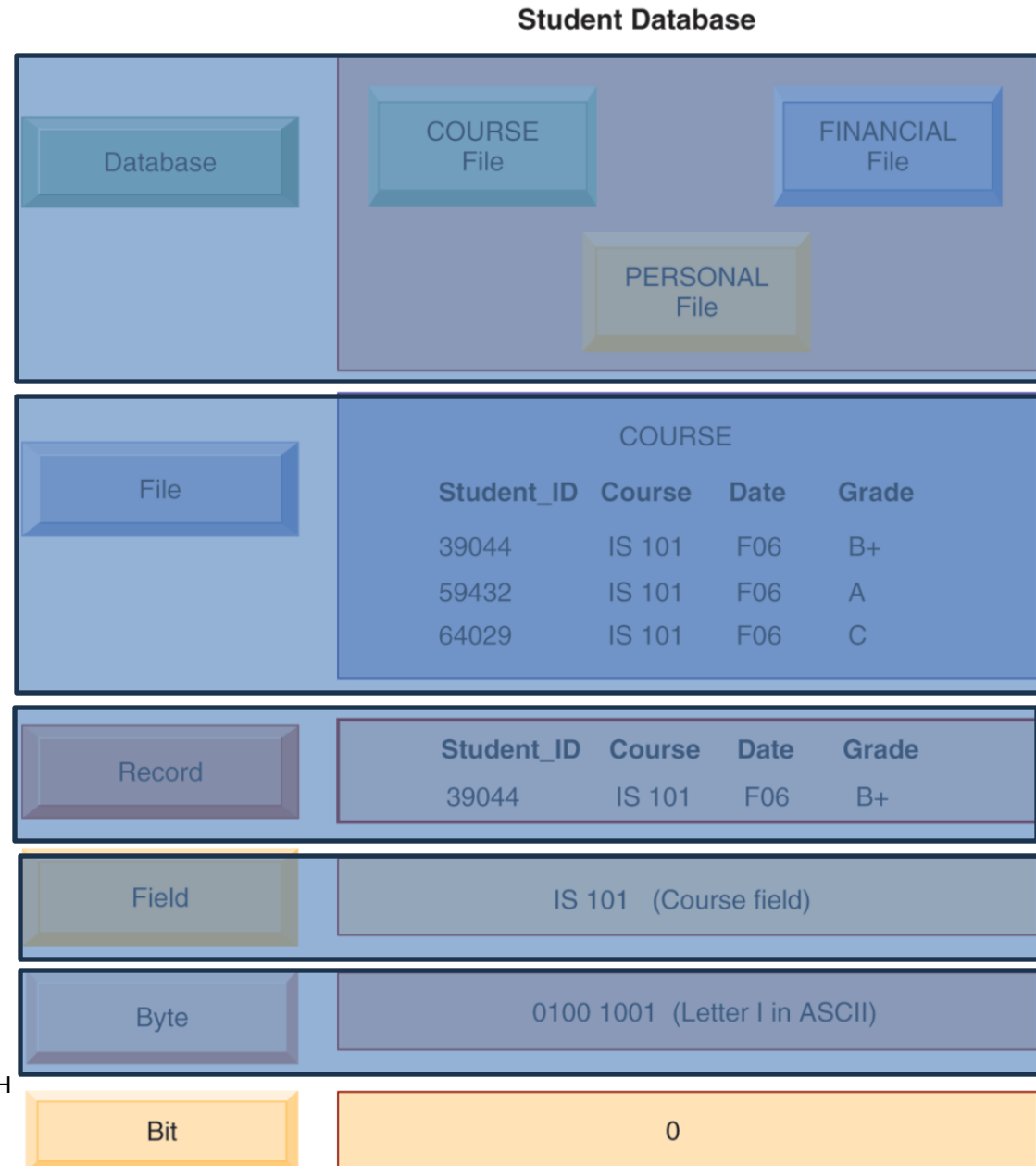
Bits can be grouped to form a **byte** to represent one character, number, or symbol.

Bytes can be grouped to form a **field**, and

Related fields can be grouped to form a **record**.

Related records can be collected to form a file, and

Related files can be organized into a **database**.



Managing Data in a Traditional File Environment

- **Problems with the traditional file environment (files maintained separately by different departments)**
 - **Data redundancy:**
 - Presence of duplicate data in multiple files
 - **Data inconsistency:**
 - Same attribute has different values
 - **Program-data dependence:**
 - When changes in program requires changes to data accessed by program
 - **Lack of flexibility**
 - **Poor security**
 - **Lack of data sharing and availability**

Relational Databases

- **Database**
 - Serves many applications by centralizing data and controlling redundant data
- **Database management system (DBMS)**
 - Interfaces between applications and physical data files
 - Separates logical and physical views of data
 - Solves problems of traditional file environment
 - Controls redundancy
 - Eliminates inconsistency
 - Uncouples programs and data
 - Enables organization to centrally managing data and data security

Relational Databases

- Key Features:
 - Structured and schema-based storage.
 - ACID compliance ensures reliability and consistency.
 - High efficiency for transactional data.
- Examples:
 - Tools: MySQL, PostgreSQL, Oracle.
 - Use Cases: Financial records, CRM systems.

HUMAN RESOURCES DATABASE WITH MULTIPLE VIEWS

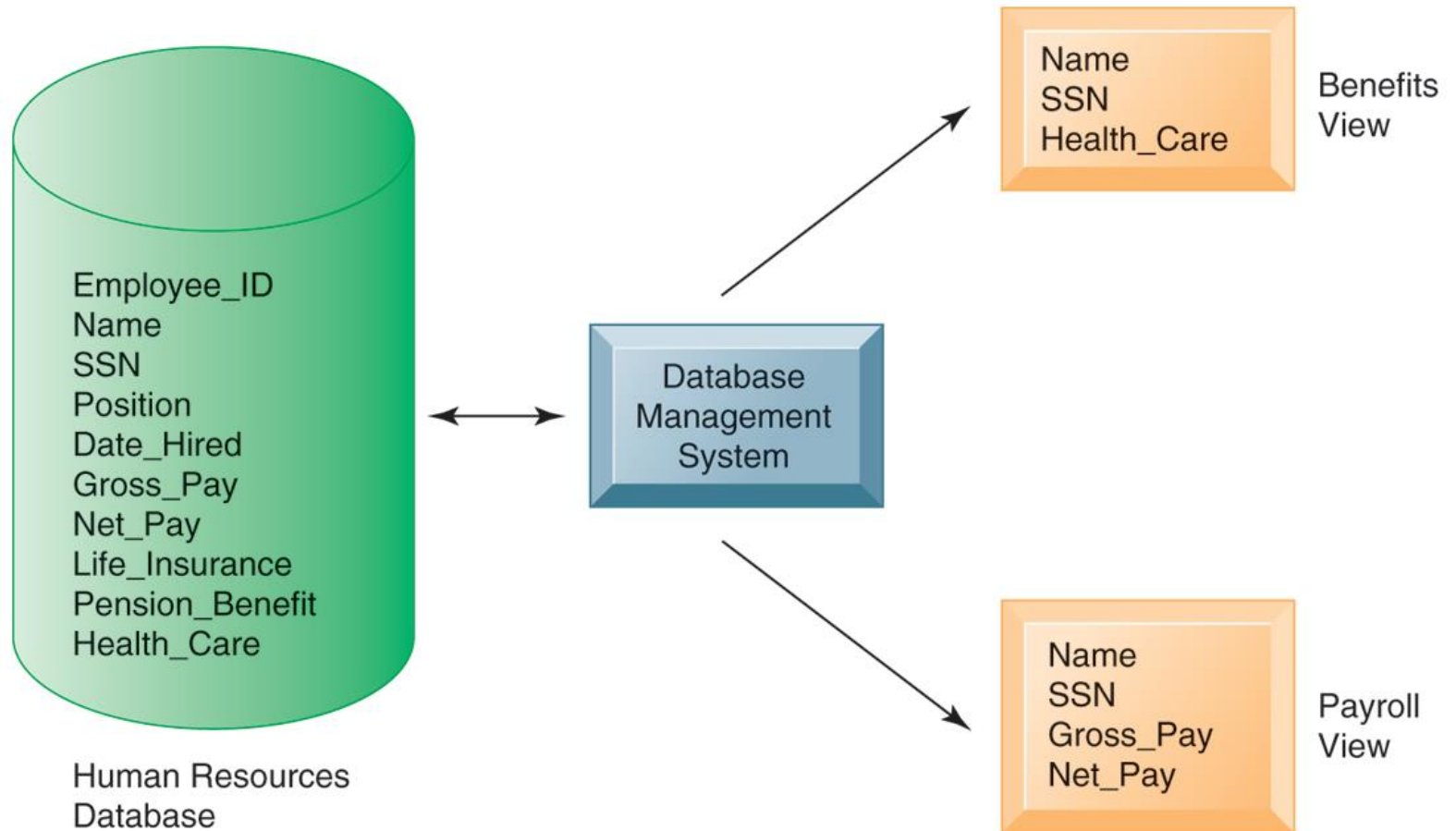


FIGURE 6-3

A single human resources database provides many different views of data, depending on the information requirements of the user. Illustrated here are two possible views, one of interest to a benefits specialist and one of interest to a member of the company's payroll department.

Capabilities of Database Management Systems (DBMSs)

- **Relational DBMS**

- Represent data as two-dimensional tables
- Each table contains data on entity and attributes

- **Table: grid of columns and rows**

- Rows (tuples): Records for different entities
- Fields (columns): Represents attribute for entity
- Key field: Field used to uniquely identify each record
- **Primary key**: Field in table used for key fields
- **Foreign key**: Primary key used in second table as look-up field to identify records from original table

Relational Database Tables

SUPPLIER

Columns (Attributes, Fields)

Supplier_Number	Supplier_Name	Supplier_Street	Supplier_City	Supplier_State	Supplier_Zip
8259	CBM Inc.	74 5 th Avenue	Dayton	OH	45220
8261	B. R. Molds	1277 Gandolly Street	Cleveland	OH	49345
8263	Jackson Composites	8233 Micklin Street	Lexington	KY	56723
8444	Bryant Corporation	4315 Mill Drive	Rochester	NY	11344

Rows
(Records,
Tuples)

Key Field
(Primary Key)

A relational database organizes data in the form of two-dimensional tables. Illustrated here are tables for the entities SUPPLIER and PART showing how they represent each entity and its attributes. Supplier Number is a primary key for the SUPPLIER table and a foreign key for the PART table.

PART

Part_Number	Part_Name	Unit_Price	Supplier_Number
137	Door latch	22.00	8259
145	Side mirror	12.00	8444
150	Door molding	6.00	8263
152	Door lock	31.00	8259
155	Compressor	54.00	8261
178	Door handle	10.00	8259

Primary Key

Foreign Key

Capabilities of Database Management Systems (DBMSs)

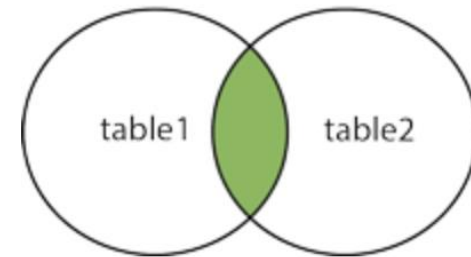
- **Capabilities of database management systems**
 - **Data definition capability:** Specifies structure of database content, used to create tables and define characteristics of fields
 - **Data dictionary:** Automated or manual file storing definitions of data elements and their characteristics
 - **Data manipulation language:** Used to add, change, delete, retrieve data from database
 - **Structured Query Language (SQL)**
 - **Many DBMS have an integrated report generation capabilities (Business Intelligence).**

Capabilities of Database Management Systems (DBMSs)

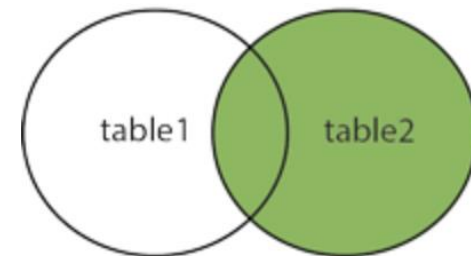
Operations of a Relational DBMS

- Three basic operations used to develop useful sets of data
 - **SELECT**: Creates subset of data of all records that meet stated criteria
 - **JOIN**: Combines relational tables to provide user with more information than available in individual tables
 - **PROJECT**: Creates subset of columns in table, creating tables with only the information specified

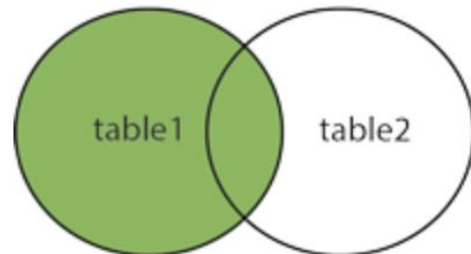
INNER JOIN



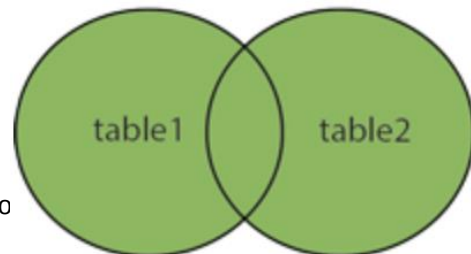
RIGHT JOIN



LEFT JOIN



FULL OUTER JOIN



THE THREE BASIC OPERATIONS OF A RELATIONAL DBMS

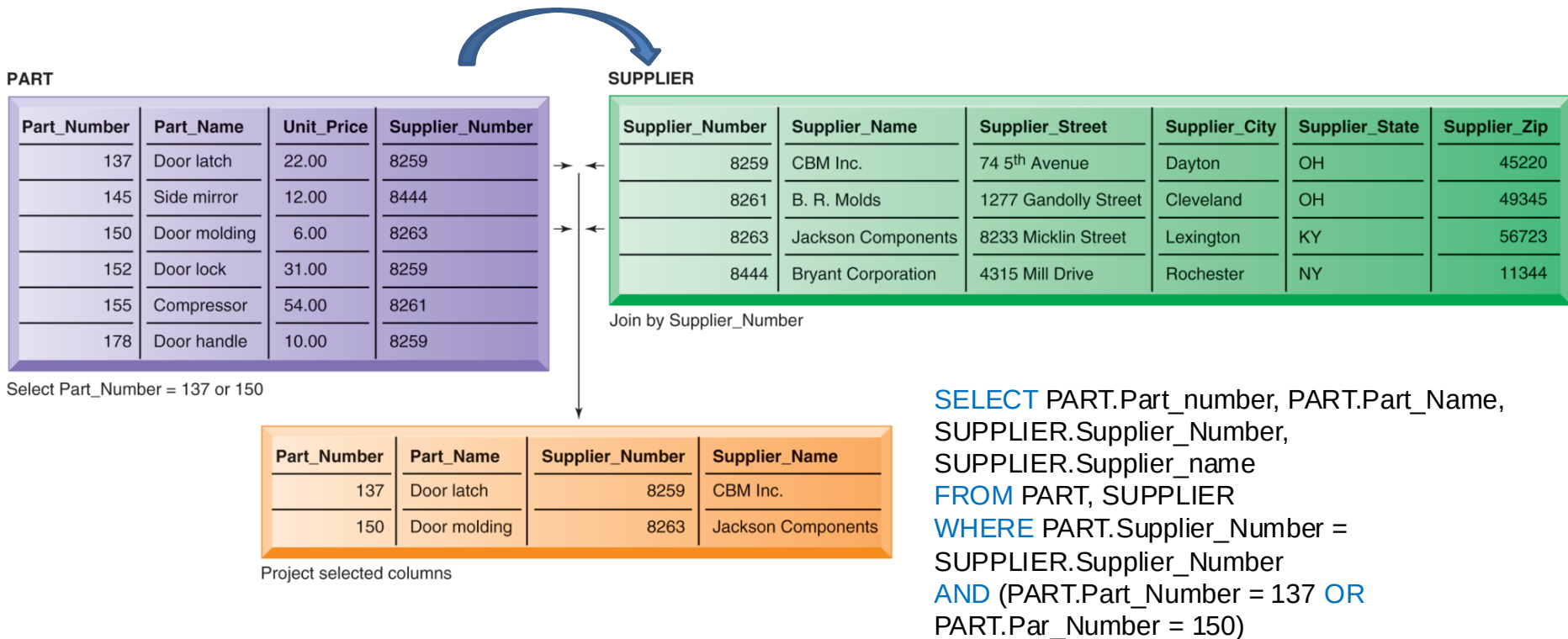


FIGURE 6-5

The select, join, and project operations enable data from two different tables to be combined and only selected attributes to be displayed.

Data management for Improving Business Decision Making

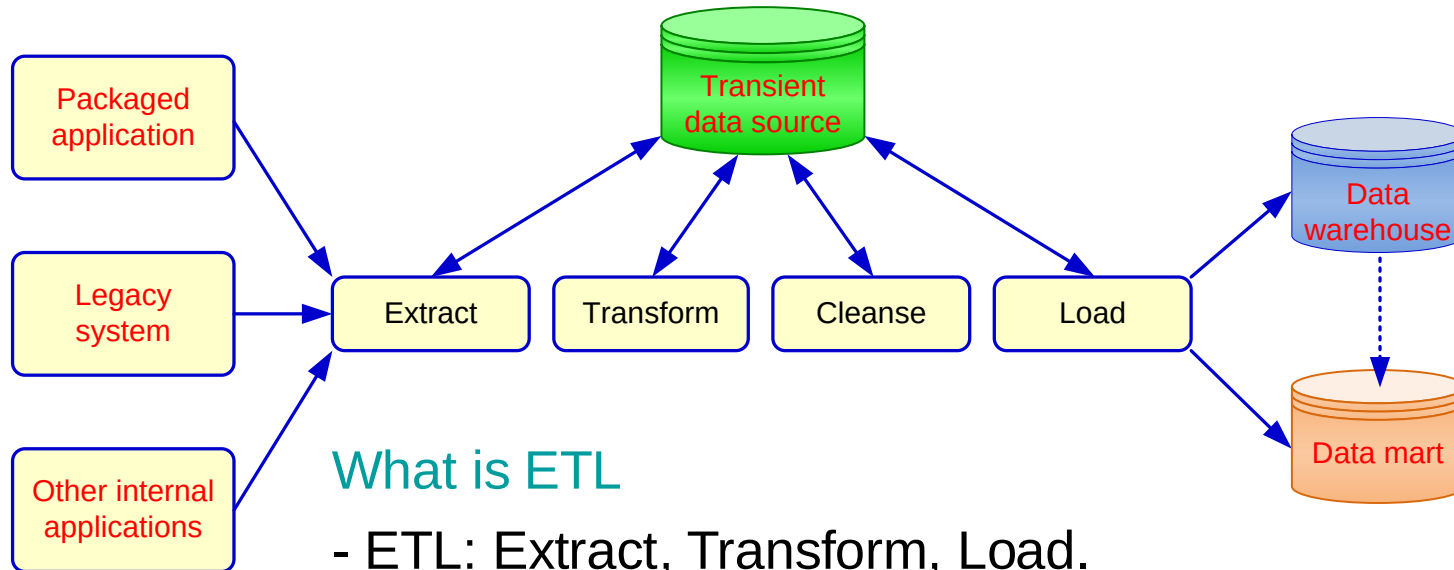
- **Data warehouse:**

- Stores current and historical data from many core operational transaction systems
- Consolidates and standardizes information for use across enterprise, but data cannot be altered
- Provides analysis and reporting tools

- **Data marts:**

- Subset of data warehouse
- Summarized or focused portion of data for use by specific population of users
- Typically focuses on single subject or line of business

Data Integration and ETL/ELT



What is ETL

- ETL: Extract, Transform, Load.

Why It's Important:

- Enables seamless data movement across systems.
- Supports analytics and decision-making.

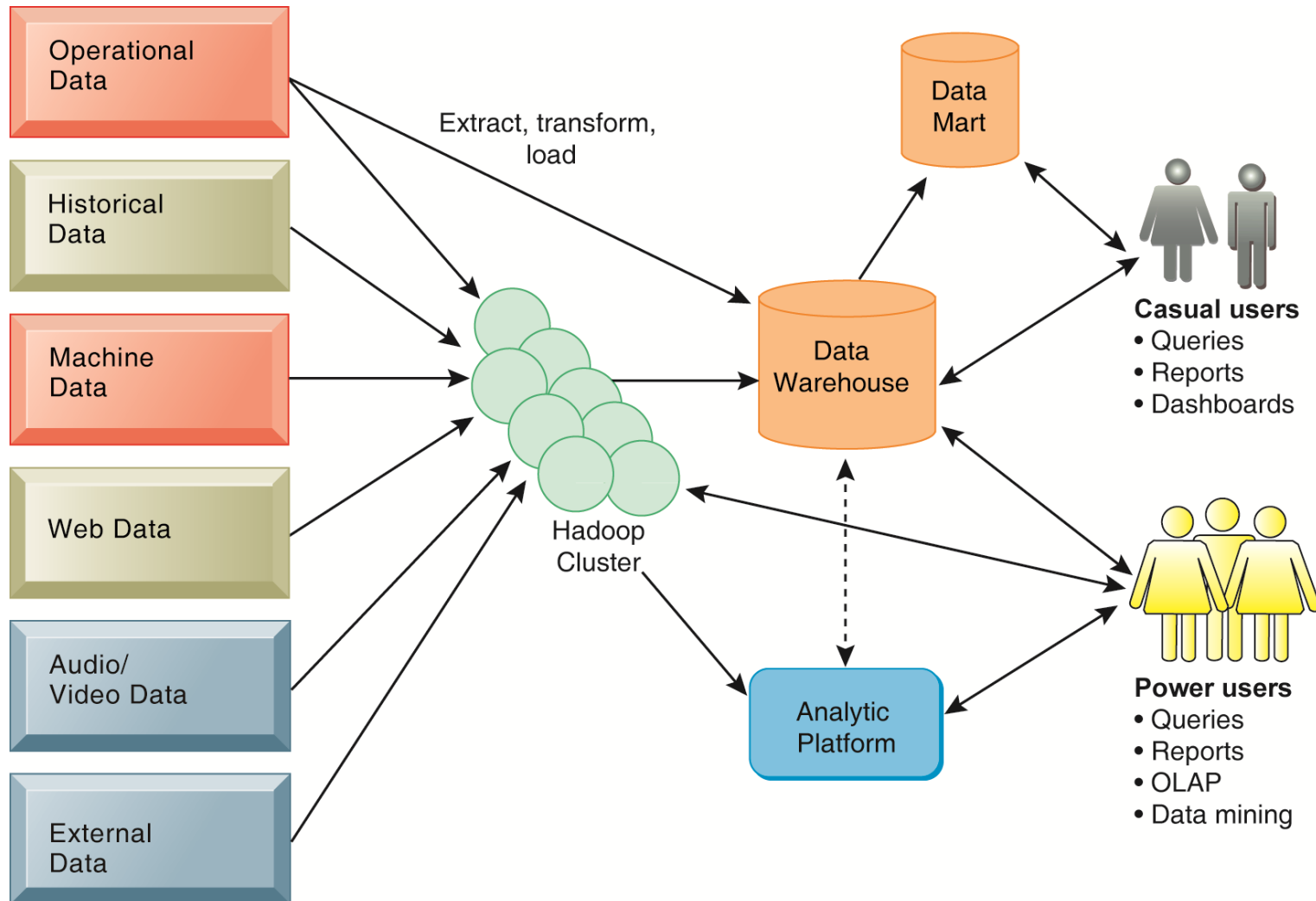
Examples:

- Tools: Talend, Apache Nifi.

Data Warehouse and Data Mart

A contemporary business intelligence infrastructure features capabilities and tools to manage and analyze large quantities and different types of data from multiple sources. Easy-to-use query and reporting tools for casual business users and more sophisticated analytical toolsets for power users are included.

FIGURE 6-12



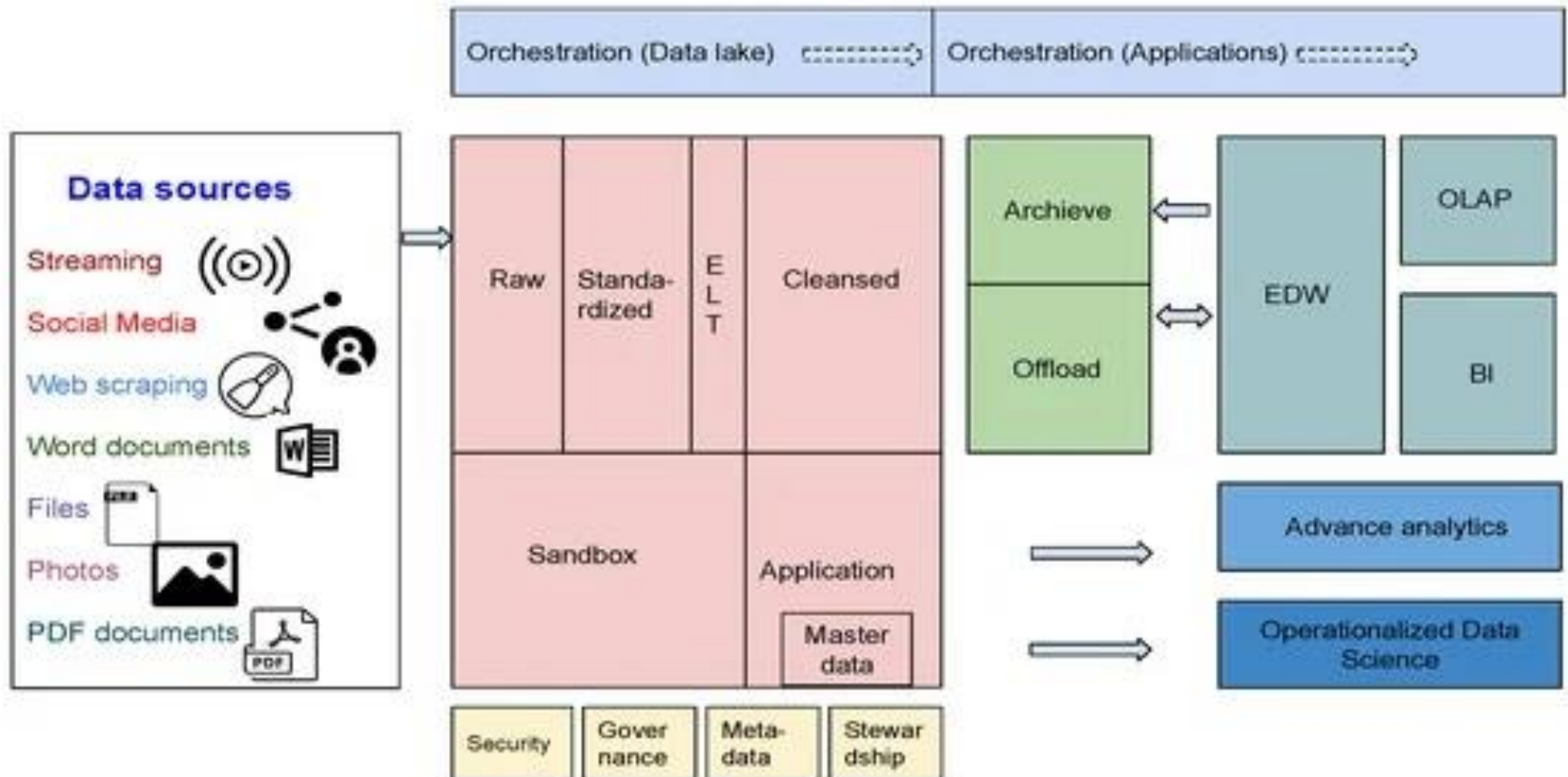
Data Warehousing and Data Lakes

Objective of a Data Warehouse: To support business intelligence, reporting, and structured analysis by providing processed, structured, and organized data optimized for querying.

Objective of a Data Lake: To enable flexibility for advanced analytics, machine learning, and exploratory data analysis by storing raw, unstructured, and semi-structured data in its native format.

Feature	Data Warehouse	Data Lake
Data Type	Structured	Unstructured
Schema	Schema-on-write	Schema-on-read
Use Case	BI and reporting	ML and AI

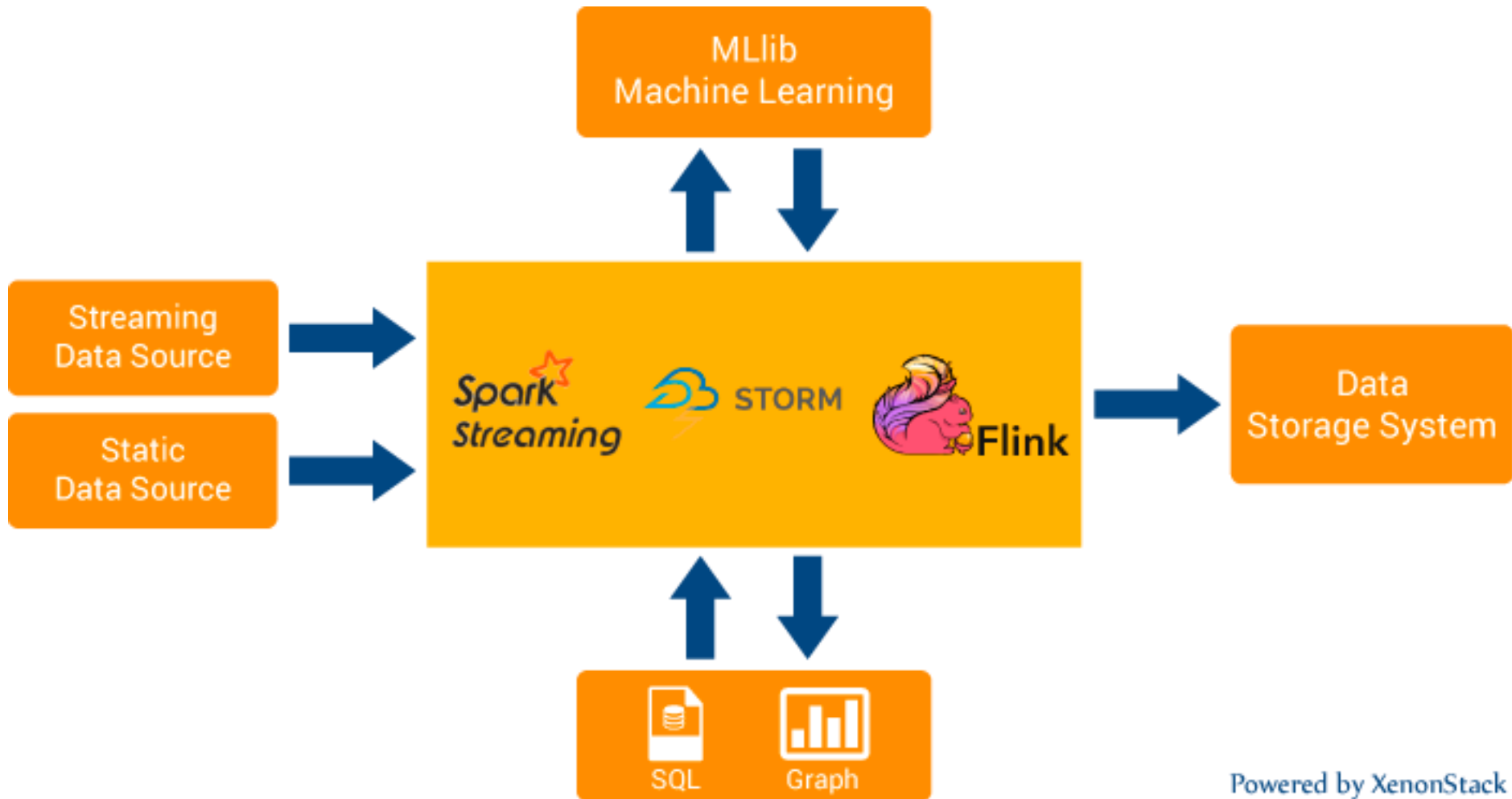
Integrated Datawarehouse and Data Lake



ADVANCED TOOLS FOR DATA MANAGEMENT

- **Non-relational databases: “NoSQL”**
 - More flexible data model (Not Only SQL)
 - Data sets stored across distributed machines
 - Easier to scale
 - Handle large volumes of unstructured and structured data (Web, social media, graphics)
 - E.g. XML, JSON, etc.
https://www.w3schools.com/js/js_json_xml.asp
- **Streaming and Real-Time Analytics**
 - Streaming and Real-Time Analytics
 - Tools: Apache Kafka, Apache Flink.
 - Use Cases: Fraud detection, supply chain monitoring.

Streaming and Real-time analytics



AI/ML Data Pipelines

- **Components:**

- Data preprocessing.
- Feature engineering.
- Model training and deployment.

- **Tools:**

- MLflow, TensorFlow Extended.

- **Example:**

- Predictive analytics for customer churn.

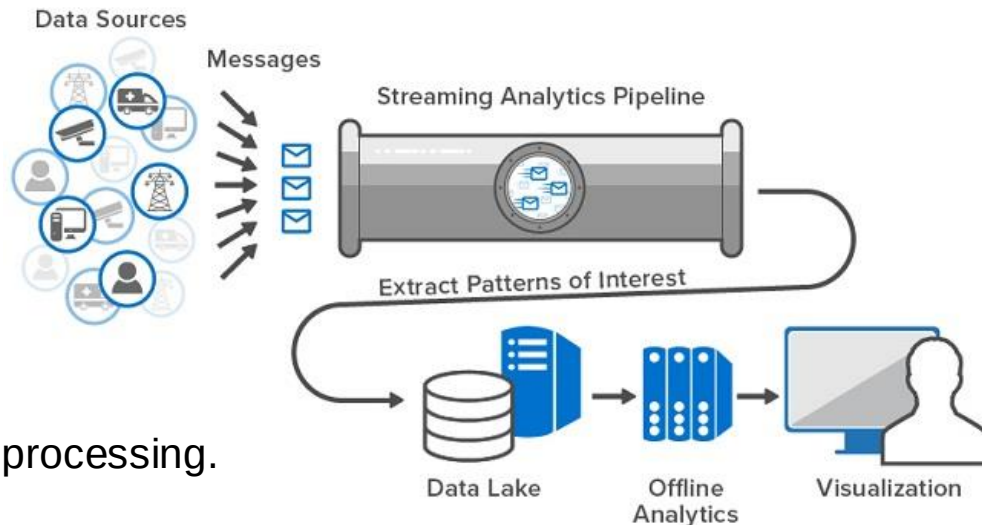


Big Data

- Enables analysis of large-scale unstructured data.
- **Key Characteristics:**
 - Volume: The sheer quantity of data being collected and stored.
 - Velocity: The speed at which data is generated and processed.
 - Variety: The different types of data sources including structured, semi-structured, and unstructured data like text, images, videos, and social media posts.
 - Veracity: The accuracy and trustworthiness of the data.
 - Value: The potential insights and business benefits that can be derived from the data.

- **Examples:**

- Tools: Hadoop, Apache Spark.
- Use Cases: IoT, social media analysis, log processing.



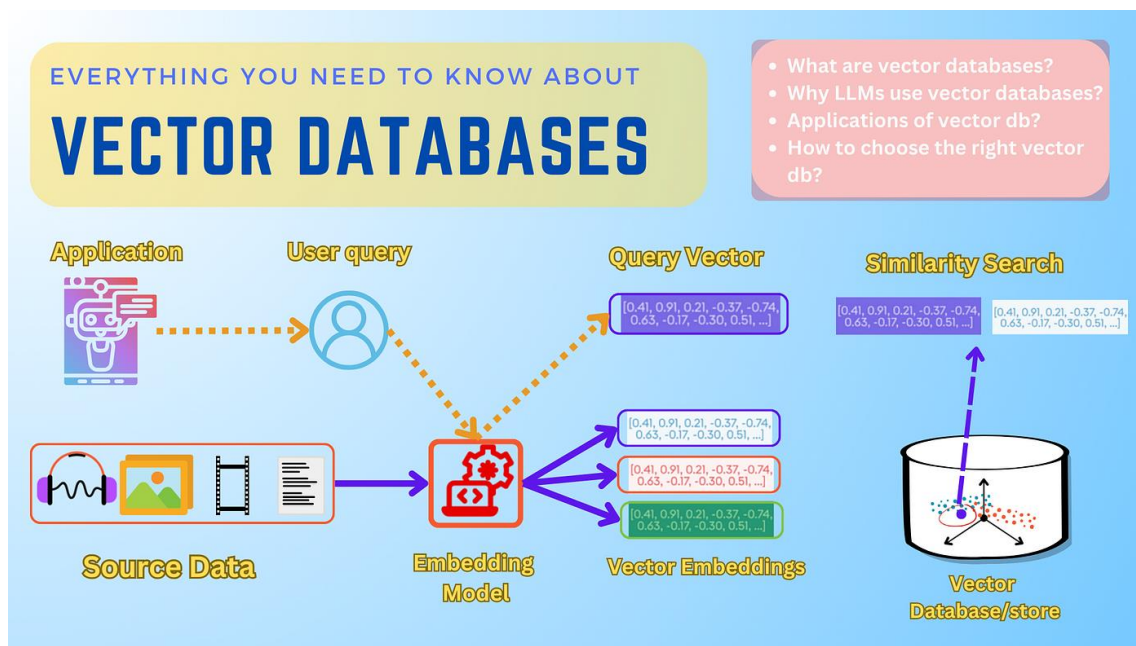
Embedding Vectors for LLMs

- **Concept:**

- Transforming unstructured data (e.g., text) into numerical vectors.
- Used for machine learning tasks like recommendation systems and semantic search.

- **Examples:**

- Tools: Sentence Transformers, Hugging Face models.
- Applications: Personalized recommendations, chatbot intelligence.

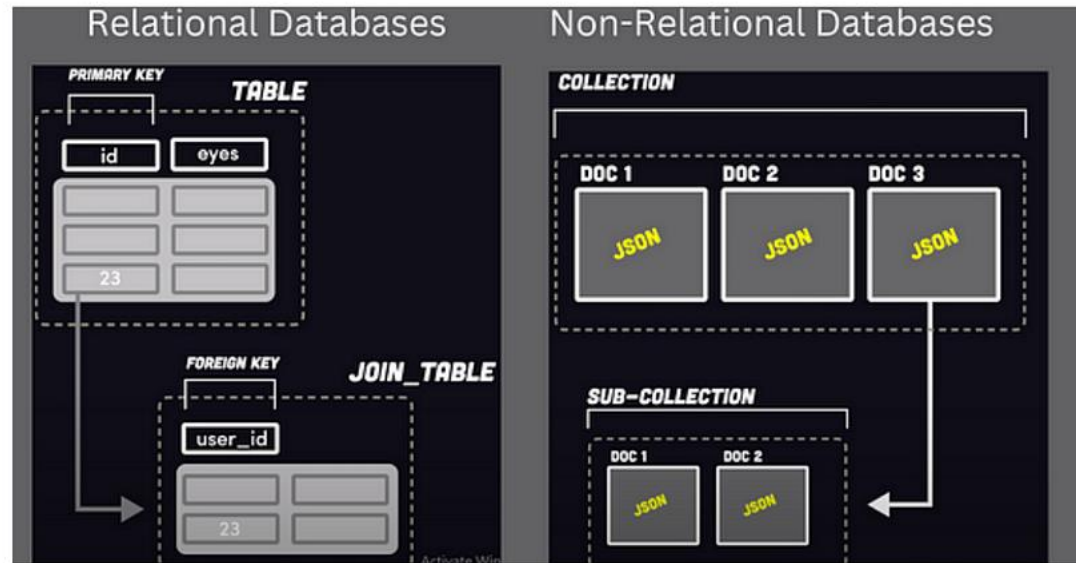


Embedding Vector –vs- Traditional Database

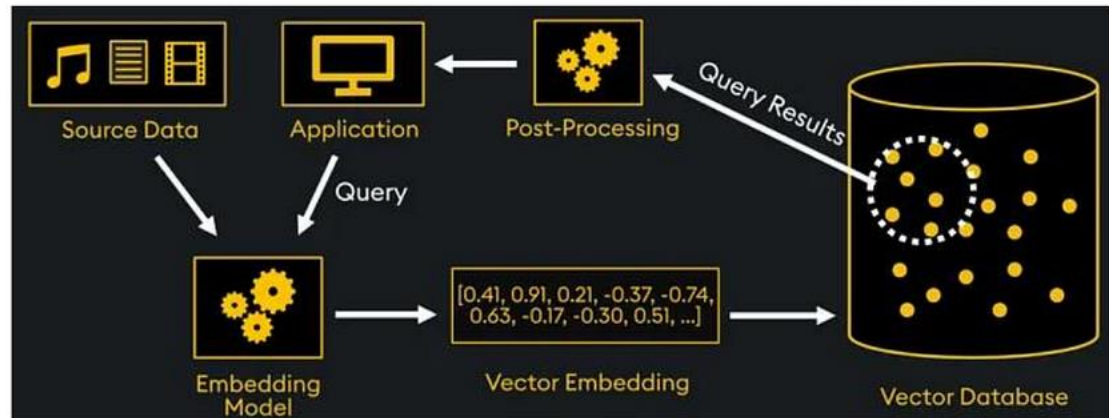
Aspects	Vector Databases	Traditional Databases
Data Representation	High-dimensional vectors capture attributes	Structured data in tables
Focus	Emphasis on similarity searches	Primarily for structured queries
Indexing	Specialized indexing structures	B-tree, hash-based indexing
Query Type	Similarity searches, content-based queries	SQL queries for structured data
Performance	Fast similarity search in high dimensions	Efficient for structured queries
Data Types	Handles various data types (images, text)	Typically suited for tabular data

Comparison of various forms of

Traditional Databases



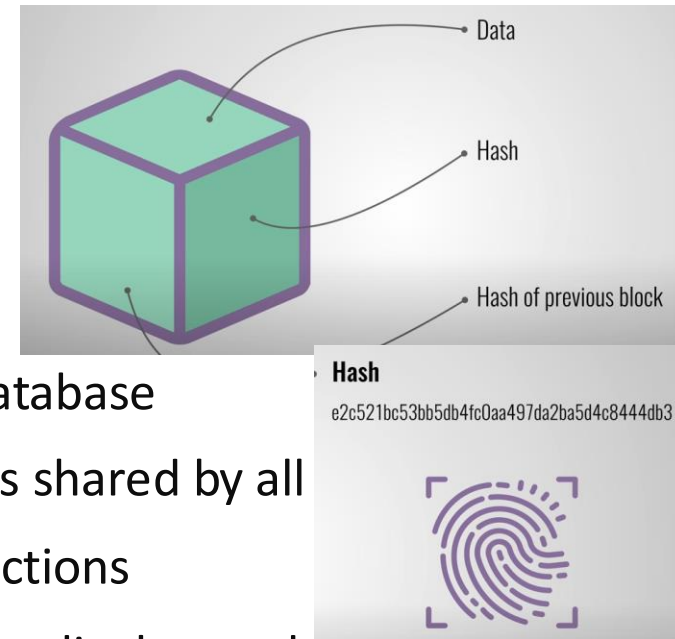
Vector Databases



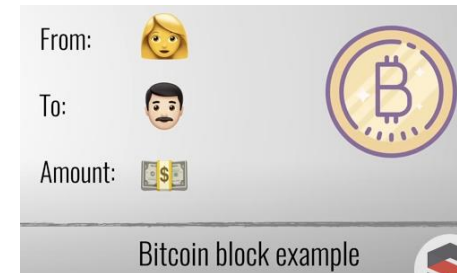
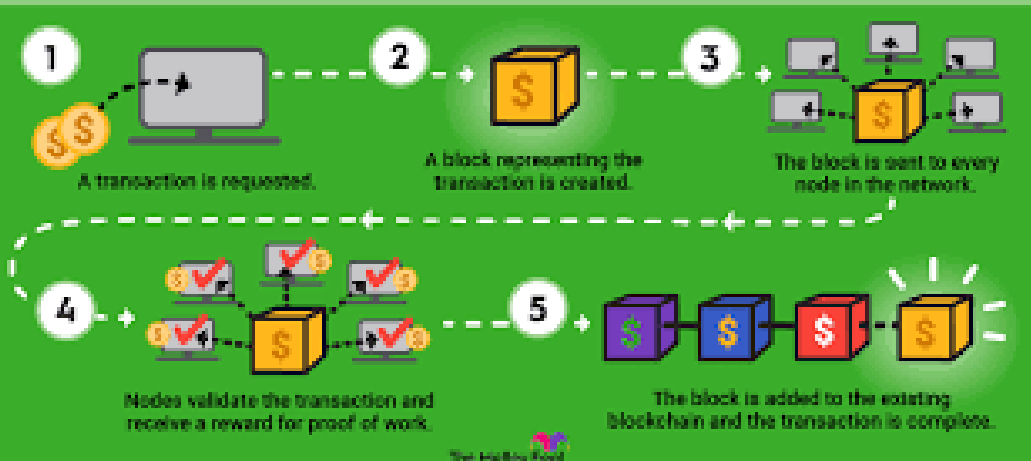
Advanced Data Management

Blockchain

- Distributed ledgers in a peer-to-peer distributed database
- Maintains a growing list of records and transactions shared by all
- Encryption used to identify participants and transactions
- Used for financial transactions, supply chain, and medical records, etc.
- Foundation of Bitcoin, and other crypto currencies



HOW BLOCKCHAIN WORKS



Smart contract

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Conclusion

- Data and analytics capabilities have changed the way businesses make decisions.
 - Companies need data-savvy professionals
 - Turn data into insights and action
- Data platform and management:
 - Evolution from a plain data file into sophisticated form of data management: File, Database, Data warehouse, Big Data, Embedding Vectors, and Block chain.